PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL
SILICON TYPES 2N7399, 2N7400, 2N7401, AND 2N7402
JANTXD , -R, JANTXVD , -R, AND JANSD, -R

This specification is approved for use by all Departments and
Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-Channel, enhancement-mode, MOSFET, radiation hardened (Total Dose and Single Event characterization - see figure 4), power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, T0-257AA.

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T 1/ T_C =+25°C	P_T 1/ T_A =+25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C =+25°C	I_{D2} T_C= +100°C	I_S 2/	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7399	50	20	100	100	±20	11.0	7.0	11.0	33	-55	N/A
2N7400			200	200		8.0	5.0	8.0	24	to	N/A
2N7401			250	250		6.0	4.0	6.0	18	+150	250
2N7402			500	500		3.0	2.0	3.0	9		500

1/ Derate linearly 0.4 W/°C for T_C > +25°C; $P_T = (T_{jmax} - T_C)/R_{\theta JC}$
2/ $I_D = ((T_{jmax} - T_C)/((R_{\theta JC})X (r_{DS(on)} \text{ at } T_{jmax})))^{1/2}$

1.4 Primary electrical characteristics at T_C = +25°C.

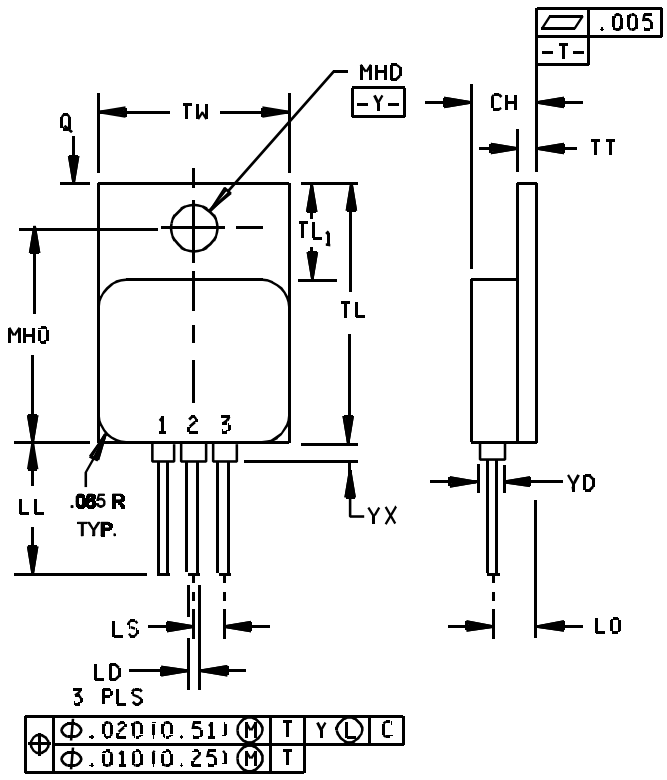
Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc	Max I_{DSS1} V_{GS} = 0 V_{GS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = 12V		$R_{\theta JC}$ Max	$I_{AS} = I_{DM}$
				T_J = 25°C at I_{D2}	T_J = 125°C at I_{D2}		
	V dc	V dc Min Max	µA dc	Ω	Ω	°C/W	A (pk)
2N7399	100	1.5 4.0	25	0.21	0.351	2.50	33
2N7400	200			0.44	0.744		24
2N7401	250			0.60	1.08		18
2N7402	500			2.70	5.37		9

1/ Pulsed (see 4.5.1).

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving *
* this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East *
* Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal *
* (DD Form 1426) appearing at the end of this document or by letter. *

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Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
CH	4.83	5.08	.190	.200
LD	0.64	0.89	.025	.035
LL	15.24	16.51	.600	.650
LO	3.05 BSC		.120 BSC	
LS	2.54 BSC		.100 BSC	
MHD	3.56	3.81	.140	.150
Q	2.88	3.37	.113	.133
TL	16.38	16.89	.645	.665
TL1	5.85	6.35	.230	.250
TT	0.89	1.14	.035	.045
TW	10.41	10.67	.410	.420
YD	1.53	2.28	.060	.090
YX		0.88		.035
Term 1	Drain			
Term 2	Source			
Term 3	Gate			

NOTES:
1. Dimensions are in millimeters.
2. Equivalents are given for general information only.
3. All terminals are isolated from case.
4. The preferred measurements used herein are the metric units. However, this transistor was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall be the rule.
5. Die to base is BeO isolated, terminals to case ceramic (AL_2O_3) isolated.

FIGURE 1. Physical dimensions for TO-257AA

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection (see 6.2).

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3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.
b. Ground test equipment, tools and personnel handling devices.
c. Do not handle devices by the leads.
d. Store devices in conductive foam or carriers.
e. Avoid use of plastic, rubber or silk in MOS areas.
f. Maintain relative humidity above 50 percent if practical.
g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.
h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table III herein. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.

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4.3 Screening (JANS, JANTX, and JANTXV). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTX and JANTXV
1/	E_{AS} test (see 4.5.4)	E_{AS} test (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
2/	Subgroup 2 of table I herein	Subgroup 2 of table I herein
9	I_{GSS}, I_{DSS1}s a minimum	Not applicable
10	MIL-STD-750, method 1042 test condition B	MIL-STD-750, method 1042 test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100% of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein.
12	MIL-STD-750, method 1042 test condition A	MIL-STD-750, method 1042 test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20% of initial value. $\Delta V_{GS(TH)1}$ = ±20% of initial value.	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20% of initial value. $\Delta V_{GS(TH)1}$ = ±20% of initial value.

1/ Shall be performed anytime before screen 10.
2/ Shall be performed after E_{AS} test, method 3161, and gate stress test.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. End point electrical measurements shall be in accordance with the applicable steps of table IV herein.

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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	1051	Test condition G, 100 cycles
3	2077	SEM
4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. 0n = 30 seconds minimum.
5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours, minimum.
5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C; t = 24 hours.
5	2037	Bond strength (Al-Au die interconnects only), Test condition A
6	3161	thermal resistance, see 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
2	1051	Test condition G, 25 cycles.
3	1042	Test condition D, 2,000 cycles, 1 cycle = 30 s min,
5 and 6	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

Subgroup	Method	Condition
2	2036	Terminal strength, test condition A, weight = 10 lbs., t = 15 sec.
6	1042	Test condition D, 6,000 cycles; 1 cycle = 30 sec. min.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF -19500 and table II herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements in figure 4.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal Resistance. Thermal Resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 2.5 °C/W. The following parameters shall apply:

a. Measuring current(I_M)..... 10 mA

b. Drain heating current (I_H)..... 1 A

c. Heating time (t_H).....Steady-state (see MIL-STD-750, method 3161)

d. Drain-source heating voltage (V_H)..... 25 V

e. Measurement time delay (t_{MD})..... 30 to 60 µs

f. Sample window time (t_{SW})..... 10 µs maximum

4.5.3 Thermal Response (ΔV_{SD} measurement). The delta V_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current (I_M)..... 10 mA

b. Drain heating current (I_H)..... 1 A

c. Heating time (t_H)..... 100 ms

d. Drain-source heating voltage (V_H)..... 25 V

e. Measurement time delay (t_{MD})..... 30 - 60 µs

f. Sample window time (t_{SW})..... 10 µs maximum

4.5.4 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$

b. L = 0.1 mH

c. $E_{AS} = 1/2\ LI_{AS}^2$

d. V_{DD} = 50 V to 150 V dc

e. Initial junction temperature = 25 °C, -5 °C, +10 °C

4.5.5 Gate stress test.

a. V_{GS} = 30 V

b. t = 250 µS, minimum

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, Bias condition C	$V_{(BR)DSS}$			
2N7399				100		V dc
2N7400				200		V dc
2N7401				250		V dc
2N7402				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	1.5	4.0	V dc
Gate current	3411	V_{GS} = +20V dc and -20 V dc, Bias condition C, V_{DS} = 0V	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80 % of rated V_{DS},	I_{DSS1}		25	µA dc
Static drain to source "ON" state resistance	3421	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$			
2N7399					0.21	Ω
2N7400					0.44	Ω
2N7401					0.60	Ω
2N7402					2.70	Ω
Static drain to source "ON" state voltage	3405	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	$V_{DS(ON)}$			
2N7399					2.43	V dc
2N7400					3.70	V dc
2N7401					3.78	V dc
2N7402					8.51	V dc
Forward voltage	4011	V_{GS} = 0V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	V_{SD}		1.8	V dc
Subgroup 3						
High temperature operation	3411	$T_C = T_J$ = +125°C	I_{GSS2}		±200	nA dc
Gate current	3413	V_{GS} = +20V dc and -20 V dc, Bias condition C, V_{DS} = 0V	I_{DSS2}		0.25	mA dc
Drain current		V_{GS} = 0V dc, Bias condition C, V_{DS} = 80% of rated V_{DS}				

See footnotes at end of table.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3-continued						
Static drain to source "ON"-state resistance	3421	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)2}$			
2N7399					0.351	Ω
2N7400					0.744	Ω
2N7401					1.08	Ω
2N7402					5.37	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	0.5		V dc
Low temperature operation		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS(TH)3}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 12 V dc, R_G = 7.5 Ω, V_{DD} = 50% of rated V_{DS} ($V_{DS} \leq$ 200 V)				
Turn-on delay time			$t_{D(on)}$			
2N7399					65	ns
2N7400					65	ns
2N7401					70	ns
2N7402					85	ns
Rise Time			t_R			
2N7399					200	ns
2N7400					160	ns
2N7401					140	ns
2N7402					120	ns
Turn-off delay time			$t_{D(off)}$			
2N7399					130	ns
2N7400					120	ns
2N7401					120	ns
2N7402					150	ns
Fall time			t_f			
2N7399					90	ns
2N7400					90	ns
2N7401					95	ns
2N7402					85	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 3 t_p = 10 ms min. V_{DS} = 80% of max. rated V_{DS} ($V_{DS} \leq$ 200 V)				

See footnotes at end of table.

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TABLE I. Group A inspection-continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 5 - continued						
Electrical measurerments		See table IV, steps 1,2,3,4,5, 6, and 7.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{G(ON)}$			
2N7399					42	nC
2N7400					42	nC
2N7401					41	nC
2N7402					36	nC
Gate to source charge			Q_{GS}			
2N7399					11	nC
2N7400					12	nC
2N7401					11	nC
2N7402					8.4	nC
Gate to drain charge			Q_{GD}			
2N7399					23	nC
2N7400					22	nC
2N7401					21	nC
2N7402					15	nC
Reverse recovery time	3473	di/dt = 100 A/µs,	t_{RR}			
2N7399		$V_{DD} \leq 30$ V				
2N7400		$I_D = I_{D1}$			280	ns
2N7401					340	ns
2N7402					420	ns
					400	ns

1/ For sampling plan, see MIL-PRF-19500.

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Table II. Group D Inspection

Inspection 1/ 2/ 3/ 4/	MIL-STD-750		Symbol	Pre-irradiation limits		Post irradiation limits		Units
	Method	Conditions		Min.	Max.	Min.	Max.	
Subgroup 1								
Not Applicable								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation (V_{GS} bias) 4/	1019	V_{GS} = 12V, V_{DS} = 0V						
Steady state total dose irradiation (V_{DS} bias)	1019	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc,	$V_{(BR)DSS}$					
2N7399		Bias condition C		100		100		V dc
2N7400				200		200		V dc
2N7401				250		250		V dc
2N7402				500		500		V dc
Gate to source Voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D =1 mA dc	$V_{GS(TH)1}$	1.5	4.0	1.5	4.0	V dc
Gate current	3411	V_{GS} = ±20V dc, V_{DS} = 0V,Bias condition C	I_{GSS1}		±100		±100	nA dc
Drain current	3413	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}, Bias condition C	I_{DSS1}		25		25	μA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12V dc, condition A, pulsed (see4.5.1), I_D = I_{D2}	$r_{DS(ON)1}$					
2N7399					0.21		0.21	Ω
2N7400					0.44		0.44	Ω
2N7401					0.60		0.60	Ω
2N7402					2.70		2.70	Ω
Static drain to source "ON"-state voltage	3405	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), I_D = I_{D1}	$V_{DS(ON)}$					
2N7399					2.43		2.43	V dc
2N7400					3.70		3.70	V dc
2N7401					3.78		3.78	V dc
2N7402					8.51		8.51	V dc

1/ For sampling plan see MIL-PRF-19500

2/ Electrical specifications are for 'D' and 'R' rad levels.

3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.

4/ At the manufacturer's option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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TABLE III. Group E inspection (all quality levels) - For qualification only

Inspection 1/, 2/, 3/, 4/, 5/, 6/	MIL-STD-750		Qualification and large lot quality conformance inspection.
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table IV, steps 1,2,3,4,5, 6, and 7	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Test condition B; 1,000 hours	
Electrical measurements		See table IV, steps 1,2,3,4,5, 6, and 7	
Steady state reverse bias	1042	Test condition A; 1,000 hours	
Electrical measurements		See table IV, steps 1,2,3,4,5, 6, and 7.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 2.5 °C/W maximum. See 4.5.2	
Subgroup 5			15 devices c = 0
Barometric pressure test (not required for $V_{BR(DSS)}$ ≤ 200V)	1001	Test condition C	
2N7401 2N7402		V_{DS} = 250 V; $I_{(ISO)}$ < 0.25 mA V_{DS} = 500 V; $I_{(ISO)}$ < 0.25 mA	

1/ See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) - For qualification only - continued.

Inspection 1/, 2/, 3/, 4/, 5/, 6/	MIL-STD-750		Qualification and large lot conformance inspection.
	Method	Conditions	
Subgroup 6			3 devices 6/ c = 0
Electrical measurements 4/		See table IV, steps 3 and 4	
SEE Effect Testing 5/		Fluence = 3e5 ± 20% ions/cm^2 Flux = 5e3 to 2e4 ions/cm^{2}sec Beam energy = 260 to 300 MeV Temperature = 25°C ± 5°C	
2N7399		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 100V and V_{GS} = -20V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 100V and V_{GS} = -10V V_{DS} = 80V and V_{GS} = -15V V_{DS} = 50V and V_{GS} = -20V	
2N7400		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 200V and V_{GS} = -20V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 200V and V_{GS} = -5V V_{DS} = 160V and V_{GS} = -10V V_{DS} = 100V and V_{GS} = -15V V_{DS} = 40V and V_{GS} = -20V	
2N7401		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 250V and V_{GS} = -15V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 250V and V_{GS} = -5V V_{DS} = 200V and V_{GS} = -10V V_{DS} = 125V and V_{GS} = -15V V_{DS} = 50V and V_{GS} = -20V	
2N7402		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 500V and V_{GS} = -15V V_{DS} = 450V and V_{GS} = -20V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 500V and V_{GS} = -5V V_{DS} = 400V and V_{GS} = -10V V_{DS} = 100V and V_{GS} = -15V	

See footnote at end of table.

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TABLE III. Group E inspection (all quality levels) - For qualification only - continued.

Inspection 1/, 2/, 3/, 4/, 5/, 6/	MIL-STD-750		Qualification and large lot conformance inspection.
	Method	Conditions	
Subgroup 6 Electrical measurements 4/		See table IV, steps 3 and 4	

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other performance specification utilizing the same die design.
3/ See Appendix A for appropriate method and test circuit.
4/ As a minimum, gate to source leakages and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table IV, with the exception of step 8, may be performed.
5/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-cm^2/mg qualify the same conditions for an LET of 26 to 30 MeV-cm^2/mg
6/ This sampling plan applies to each bias condition specified.

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TABLE IV. Group A, B, C and E electrical measurements

Step	Inspection 1/, 2/, 3/	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, Bias condition C	$V_{(BR)DSS}$			
	2N7399				100		V dc
	2N7400				200		V dc
	2N7401				250		V dc
	2N7402				500		V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	1.5	4.0	V dc
3.	Gate current	3411	V_{GS} = +20V and -20V dc, Bias condition C, V_{DS} = 0V	I_{GSS1}		±100	nA dc
4.	Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80% of rated V_{DS}	I_{DSS1}		25	μA dc
5.	Static drain to source "ON"-state resistance	3421	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$			
	2N7399					0.21	Ω
	2N7400					0.44	Ω
	2N7401					0.60	Ω
	2N7402					2.70	Ω
6.	Static drain to source "ON"-state voltage	3405	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	$V_{DS(ON)}$			
	2N7399					2.43	V dc
	2N7400					3.70	V dc
	2N7401					3.78	V dc
	2N7402					8.51	V dc
7.	Forward voltage	4011	V_{GS} = 0V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	V_{SD}		1.8	V dc
8.	Thermal response	3161	See 4.5.3	ΔV_{SD}		90	mV

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 4, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.
c. Subgroup 5, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

(See notes continued next page)

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TABLE IV. Group A, B, C and E electrical measurements - continued.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2 and 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.
b. Subgroup 6, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

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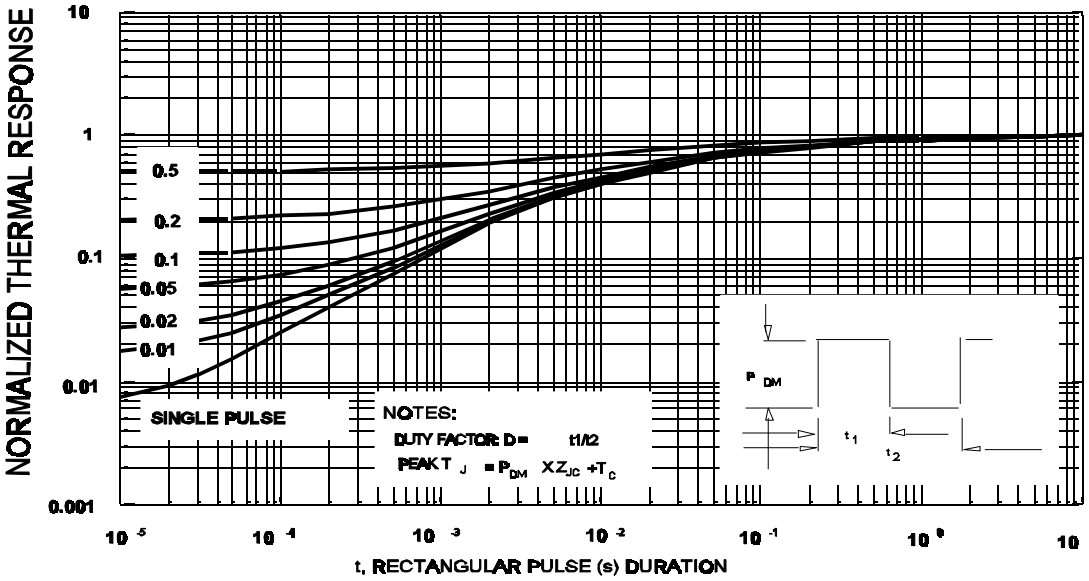


FIGURE 2. Thermal response curves

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2N7399

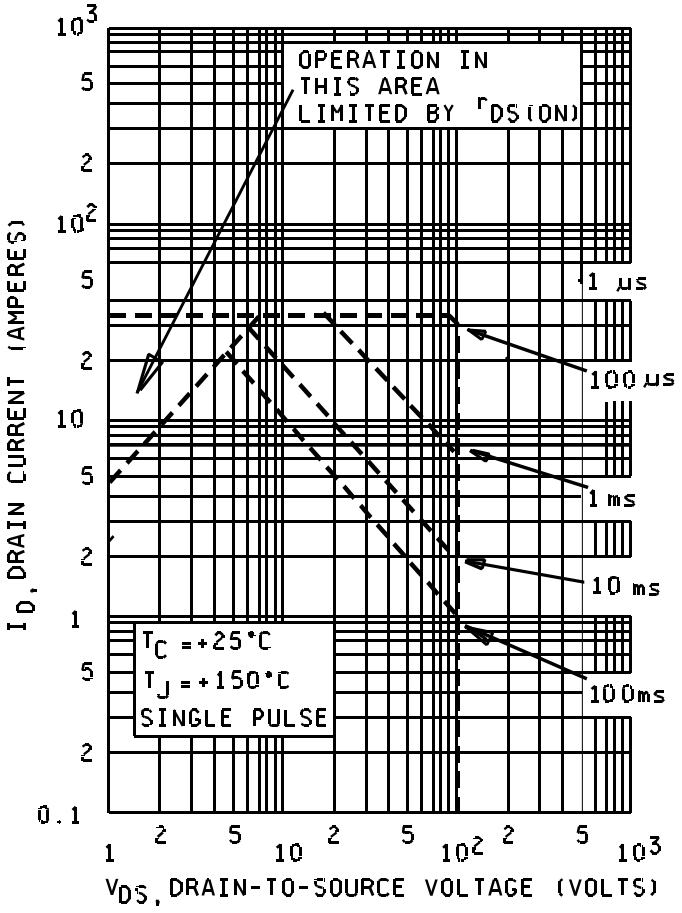


FIGURE 3. Safe operating area graphs

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2N7400

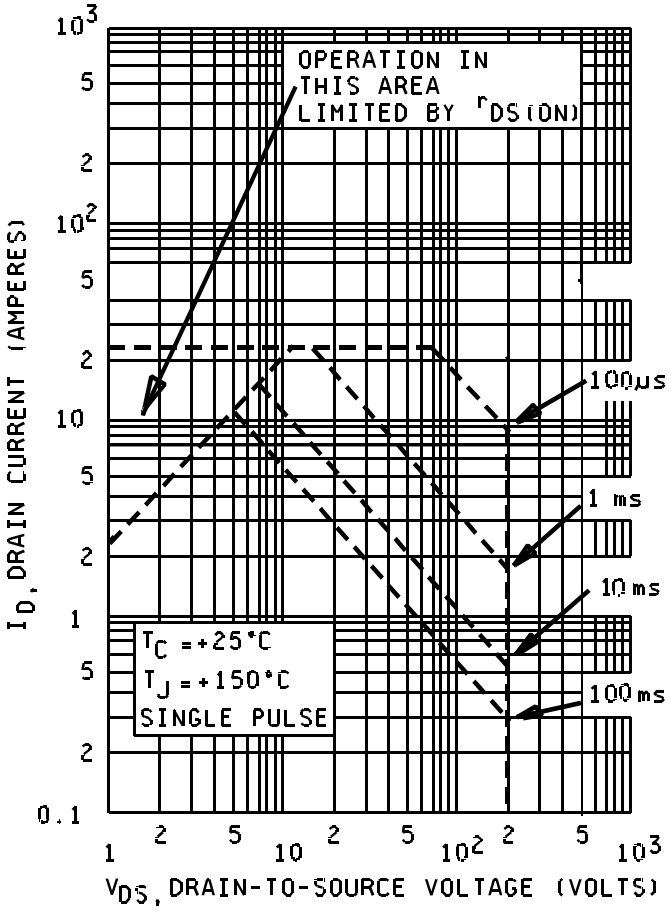


FIGURE 3. Safe operating area graphs - Continued

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2N7401

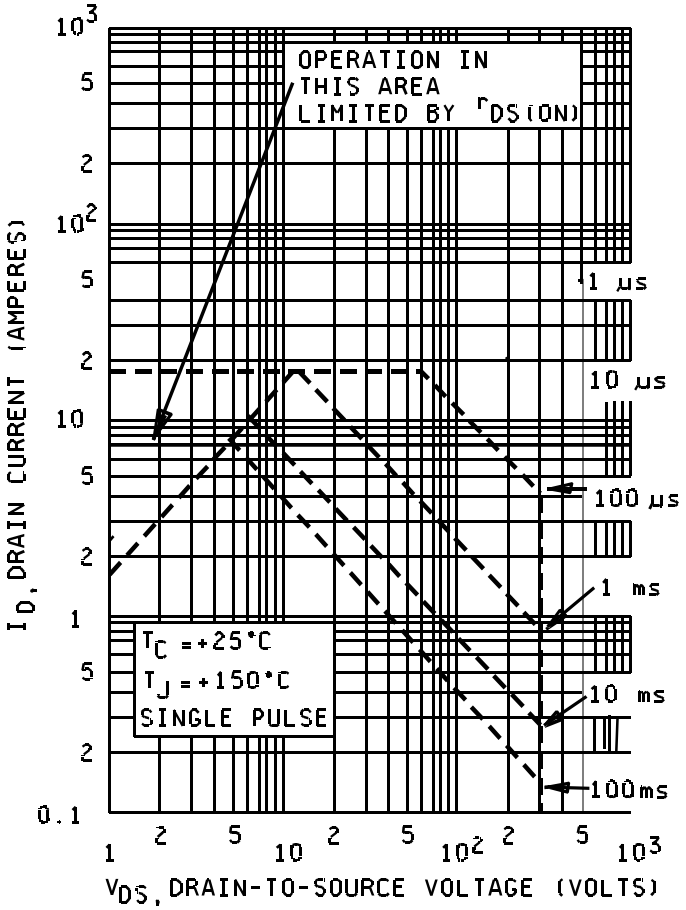


FIGURE 3. Safe operating area graphs - Continued

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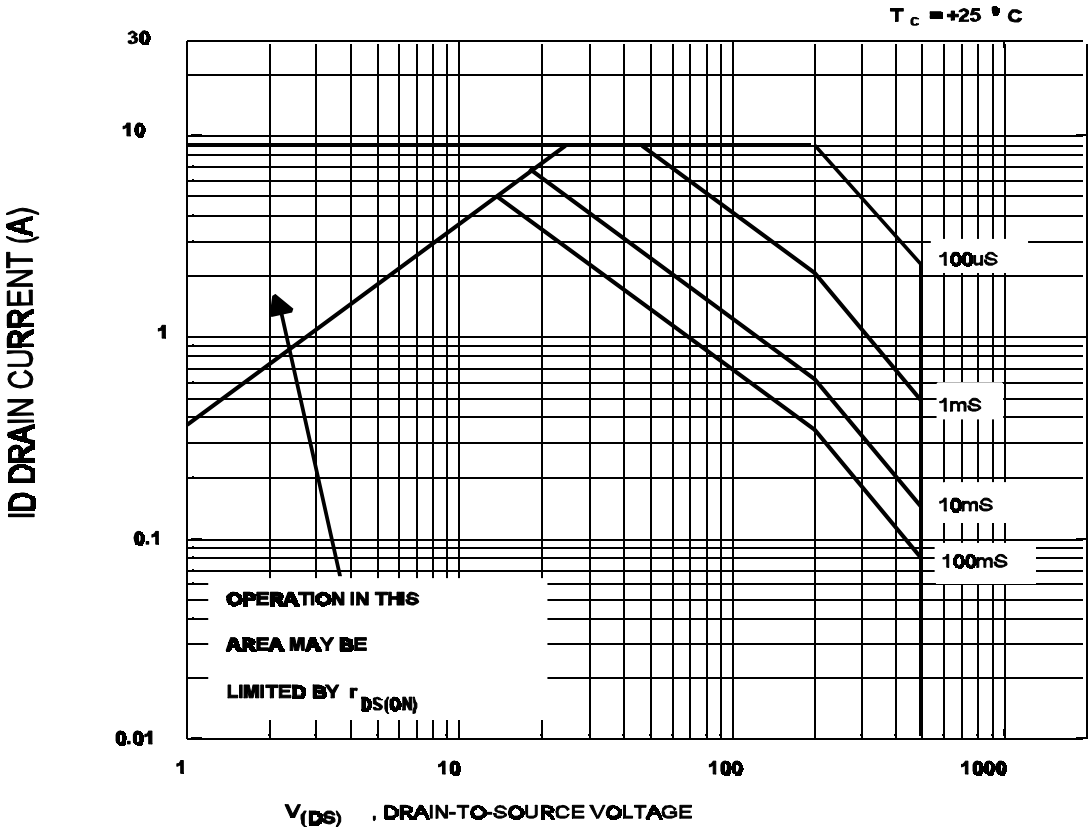


FIGURE 3. Safe operating area graphs - Continued

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2N7399

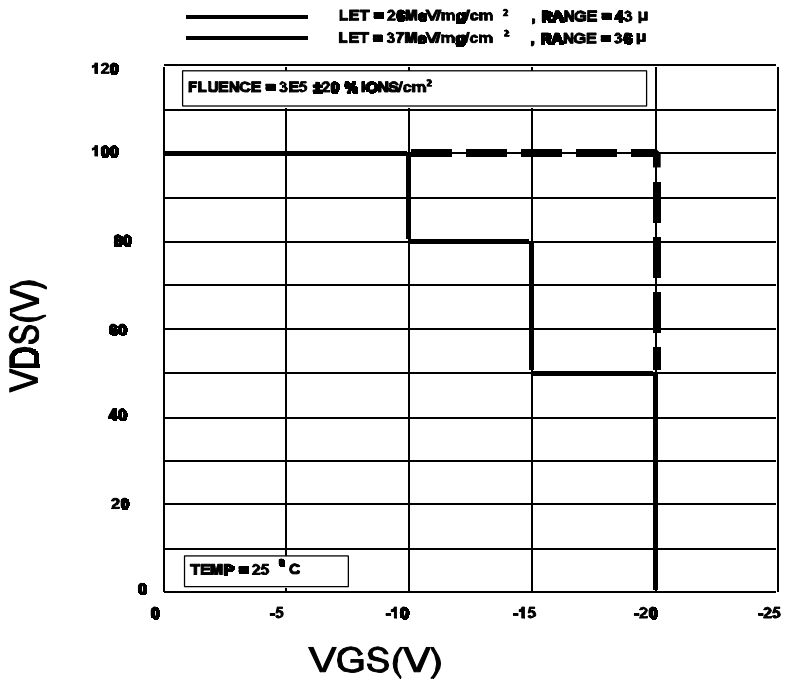


FIGURE 4. Single event effects safe operating area graphs

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2N7400

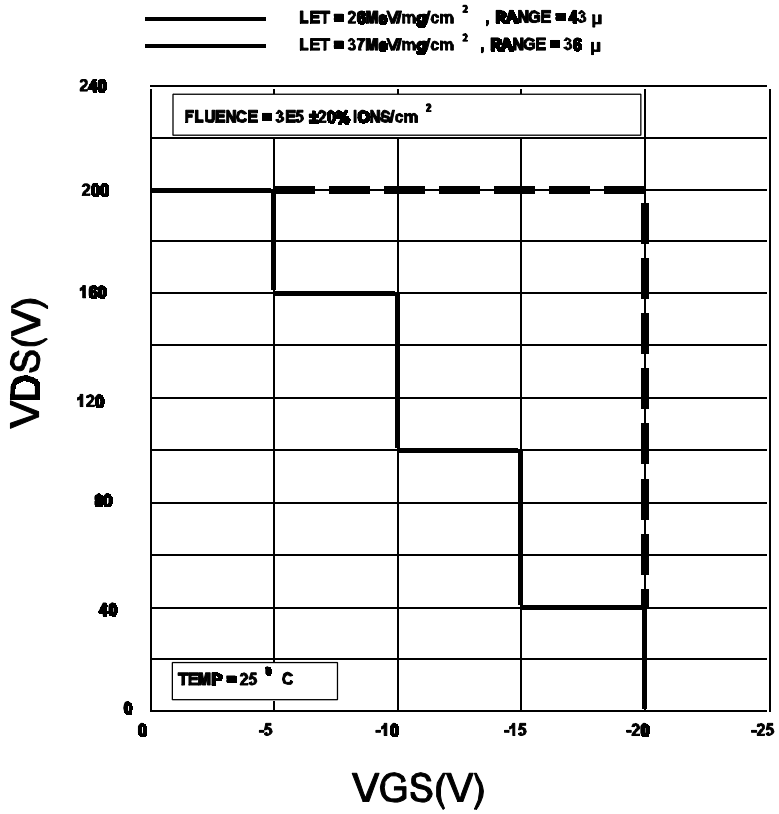


FIGURE 4. Single event effects safe operating area graphs - Continued

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2N7401

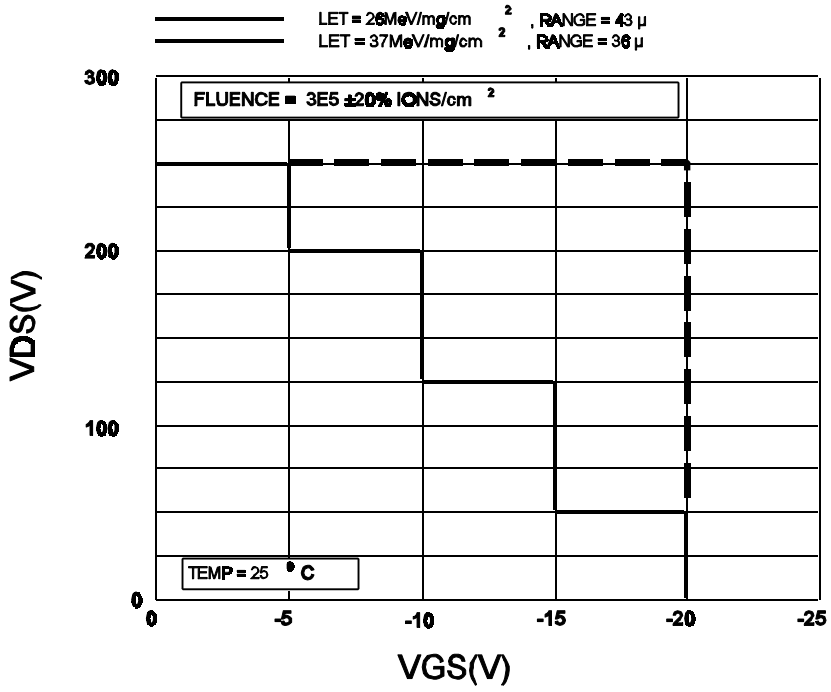


FIGURE 4. Single event effects safe operating area graphs - Continued

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2N7402

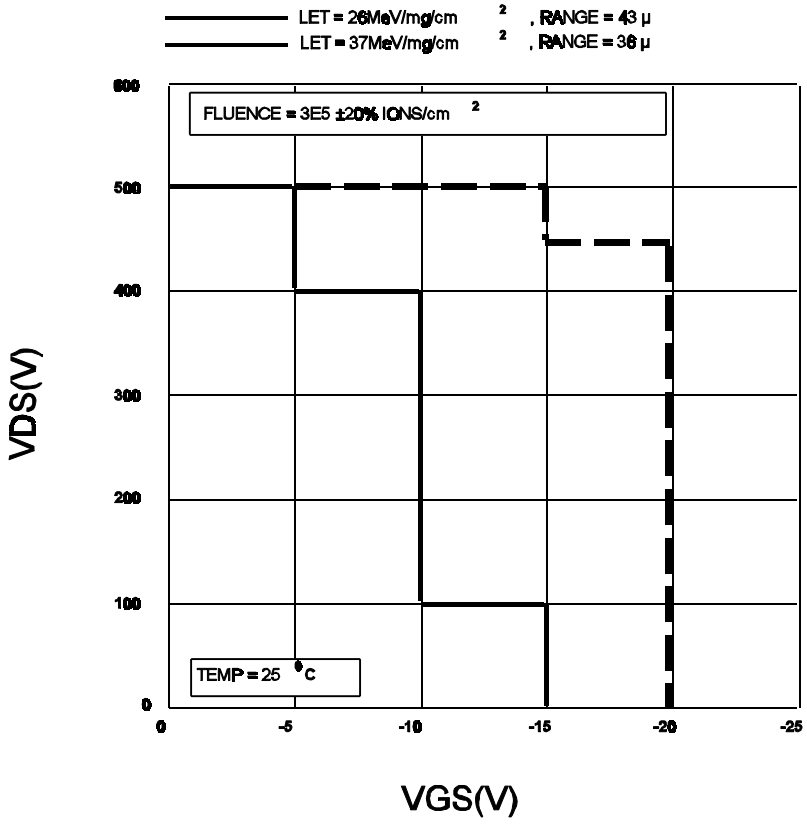


FIGURE 4. Single event effects safe operating area graphs - Continued

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of specification.

b. Issue of DODISS to be cited in the solicitation., title, number and date of specification.

c. Lead finish or formation may be specified (see 3.3.1).

d. Type designation and product assurance level.

6.3 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

Generic P/N	Military P/N
SS130	2N7399
SS230	2N7400
SS234	2N7401
SS430	2N7402

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43210.

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APPENDIX A

1 Single Event Gate Rupture and Drain Burnout Tests

1.1 Scope. This test method defines the requirements for qualification testing of discrete power FETs, for single-event gate rupture (SEGR) and single-event drain burnout (SEB) resulting from energetic heavy ion irradiation from an accelerator. This procedure can be used with either steady state or pulse type irradiations. This test is destructive, it requires delidding of devices for most sources and is likely to result in device degradation or destruction.

1.2 SEGR/SEB Screening. All testing requires a chosen drain to source capacitor sufficiently large to damage the device in the event that SEB occurs. Accordingly, there must be no significant limiting drain or source resistance as is commonly used to obtain an SEB “error count” for past practices. Accordingly, SEB becomes a destructive test just as it is in a normal application. Although there will sometimes be confusion about the nature of the failure, (SEGR or SEB), it is moot.

1.2.2 Assurance testing. Assurance testing requires a chosen sample size and number of acceptable failures. Assurance testing requires a fixing of V_{DS}, V_{GS}, LET, energy, fluence, an approximate flux, angle of incidence (zero, which is the most severe), and temperature (SEGR is independent of temperature and SEB is slightly more severe at low temperature).

1.3 Apparatus

1.3.1 Power Supply. Using the test circuit of figure 5, regulated DC power supply with floating output to produce the voltages required to bias the junction.

1.3.2 Oscilloscope. Either a single dual-beam, or two single-beam oscilloscopes that have adequate bandwidth capability to ensure that radiation response and peak steady-state values are accurately displayed.

1.3.3 Oscilloscope Camera(s) and Film. Oscilloscope Camera(s) and Film capable of recording single transient traces at a sweep rate consistent with good resolution at the pulse width used in the test.

1.3.4 Digitizers. Digitizers with Bandwidth, Sampling Interval and Time-base Capabilities adequate for handling the transient signals with good resolution for all pulse widths utilized in the test may be used. Hard copy printouts of the recorded signal may be a part of the capability of this apparatus.

1.3.5 Cabling. Cabling to complete the connection of the test circuit in the exposure area with the power supply and oscilloscopes in the data area. Any type of ungrounded wiring may be used to connect the power supply to the bias points of the test circuit; however, coaxial cables properly terminated at the oscilloscope input are required for the signal leads.

1.4 Sampling. This is a destructive test. Therefore, for the purposes of lot testing or lot qualification, a representative sample should be drawn from the lot in accordance with a test plan that is agreed upon by parties to test. Any suitable package may be used for this test.

1.5 SEGR Test Plan A test plan shall be developed which specifies the following:

a. Devices to be tested and description.
b. Sample selection process and number of samples.
c. Bias and ambient conditions.
d. Bias step increments.
e. Dosimetry.
f. Method of monitoring the DUTs.
g. Temperature monitoring.
h. Beam condition including ion species to be utilized in performing the tests.
i. Data recording requirements.
j. Preliminary analysis, if required to establish a theoretical maximum V_{DS}-LET.

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1.6 Preparation of Apparatus The following steps should be followed in the preparation of the apparatus:

a. Utilize the appropriate test circuit (see figure 5).

b. Perform the dosimetry calibration and record.

c. Verify the response of the test circuit and associated cables:

d. Connect the device tester to the DUT.

e. Adjust V_{DS} to a value V_{DS} = 0.

1.6.1 Beam Tuning. The beam shall be adjusted to provide the required uniformity, level of contamination, and energy.

1.6.2 DUT Preparation.

a. Verify the electrical performance of the DUT.

b. Delid the device.

c. Any organic overcoats must be removed before testing is initiated.

d. Reverification of the electrical performance of device after delidding.

1.7 Device Screening.

a. Mount the DUT in the test circuit and place the assembly into the vacuum chamber.

b. Measure and record the ambient temperature in the vicinity of the DUT.

c. Verify the beam tuning.

d. Apply bias to the DUT as defined in the test plan.

e. Irradiate the DUT with the selected heavy ion.

f. Observe the oscilloscope or equivalent output measurement to verify DUT's operation.

g. Reverification of the electrical performance of device after ion exposure.

1.8 Test Reporting. The test data sheet shall contain the following information:

a. Dates, times, names of test personnel.
b. Type of accelerator, name and location; ion species and energy.
c. DUT types, serial numbers, functional description, technology, manufacturer, date code and mask number if known.
d. Device duty factor and fractional portion of the chip tested, if applicable.
e. Purpose for each test run and any changes from previous test run.
f. DUT operating parameters (bias, clock frequency, temperature, etc)
g. DUT test patterns or operational modes, including duty factor.
h. Beam angle.
i. Particle count (related to fluence), run time.
j. Number of errors, locations, and special comments (anomalous incidents).
k. Transient events and recovery time when instrumental.
l. Special test results; e.g., SEB, SEGR, etc.

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1.8.1. The test report shall contain the following information:

a. Test objective

b. Complete description of the product(s) tested.

c. Description of test setup and methods.

d. Complete description of bias and ambient conditions.

e. Summary of results.

f. Conclusions.

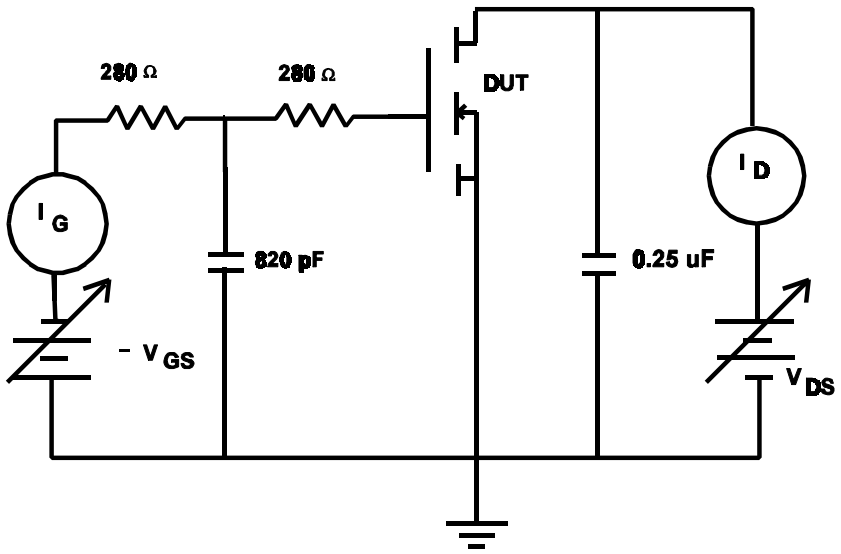


FIGURE 5. SEGR test circuit.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Preparing activity:
DLA - ES

(Project 5961-1844)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/632	**2. DOCUMENT DATE (YYMMDD)** 96/11/29

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTORS, N-CHANNEL SILICON TYPES 2N7399, 2N7400, 2N7401, AND 2N7402 JANTXD , -R, JANTXVD , -R, AND JANSD, -R

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0516
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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